

Tunisia, audited — investor economics

The FX bridge by leg, per-pathway DSCR computed in code, the capital stack, and the exact signatures that move “commercially validated” from zero.

Provenance law: MEASURED = cited anchor · DERIVED = arithmetic on measured values · ASSUMED = declared default. commercially-validated = 0 — no buyer has signed anything; published as zero on purpose. Generated 2026-09-09 by nations_audit.py / make_nations_pdfs.py — the same code that renders the page.

1. The FX bridge, tiered by confidence

LEG	\$M/YR	GRADE
Strategic metals (U + heavy REE, acid stream)	\$75-86M	STRONG
Solar fuel substitution (500 MW)	\$95-116M	PLAUSIBLE
Hydrogen → ammonia (loop-internal offtake)	\$7-25M	<i>SPECULATIVE</i>
Terfas (desert truffles)	\$14-29M	PLAUSIBLE
Food import substitution	\$30-30M	<i>SPECULATIVE</i>
Compute colocation services	\$15-15M	<i>SPECULATIVE</i>
ESG financing delta	\$8-12M	PLAUSIBLE
Total (mid)	~\$279M/yr	tiered by grade below

\$80M carries a proven flowsheet or a cited anchor; \$137M is gated; \$61M rests on assumed volume or price. Zero dollars are commercially validated — the first buyer moves the ledger.

1.1 The Fed transmission — measured, not restated

World Bank API, measured 2026-09-09 (Pg1): external debt **\$40.46B**, **78.7% of GDP**; debt service **26.2% of export revenues**; reserves **\$9.34B** = 4.3 months of imports; food \$3.00B + fuel \$4.90B of imports; CA -1.5% of GDP; growth 2.5%, inflation 5.2% (2025).

Channel 1 — the debt trap: the “original sin” (Pg12) — no long-term borrowing in the dinar abroad — exposes the state to the Fed cycle (Pg13): Fed hikes lift global yields, refinancing turns expensive, and the 26.2% of export revenues that debt service absorbs compounds while reserves (4.3 months) drain.

Channel 2 — imported inflation: the rate differential pulls capital to US safe assets, the dinar depreciates, and the dollar invoice on food (\$3.00B) and fuel (\$4.90B) inflates. The BCT — autonomous, non-monetizing under Organic Law 2016-35 (Pg4/Pg5) — must hold rates to defend the dinar: correct orthodoxy, strangled growth.

The loop’s answer: owned export engines earn ~\$117M/yr of gross exports outside the Fed cycle; the substitution legs take \$152M/yr out of the dollar invoice directly; reserves rise, the rating runs B-/Caa1 →

BB-, and the refinancing premium shrinks for the state with its own hard-currency engines. Not defiance of the Fed — independence from the channel.

The macro context is the thesis: the same Fed transmission that sets the emerging-market cycle prices this state's debt, and the loop's engines are the only line item in the country that earns hard currency *outside* that channel. That is why the FX bridge above is the investor's first table, not an appendix.

2. Pathway economics — computed in code (nations_v5.py)

P2 • Strategic metals (acid-stream U + heavy REE)

LINE	VALUE
Revenue	\$75-86M/yr (716.5k lbs U&sb3;O&sb8; @ \$80-95/lb + ~585 t REO)
OPEX	\$54-74M/yr (SX \$59-83/lb + shared circuit)
Gross margin	0-37%
CAPEX / payback	~\$100M / 6.3 yr (mid)
DSCR (60% debt @7%/15y)	2.40 — passes at contract prices
Gate	pilot U recovery ≥85% · all-in ≤\$75/lb · offtake signed

P3 • Solar IPP (500 MW) — the honest finding

LINE	VALUE
Revenue / OPEX	\$27.2M / \$8M (500 MW × 20% CF × \$31/MWh)
CAPEX	~\$550M
DSCR @8%/20y	0.46 — fails
DSCR @4.5% DFI/25y	0.69 — fails
Bankable region (DSCR ≥1.15)	\$0.70/W + 21% CF · \$0.70/W + 22% CF · \$0.70/W + 23% CF · \$0.70/W + 24% CF · \$0.80/W + 24% CF
Clearing PPA	≥\$45652/MWh at mid-case capex/CF
Verdict	bankable ONLY lean, inside the region, or with grant/sovereign support — published, not hidden

P4 • Municipal desalination (150,000 m³/d)

LINE	VALUE
Revenue	\$30.1M @\$0.55 → \$38.3M @\$0.70 blended
OPEX / CAPEX	\$24.6M (@\$0.45/m ³) / ~\$165M
DSCR @\$0.55	0.54 — fails
DSCR @\$0.70	1.36 — clears

LINE	VALUE
Gate	≥60% contracted offtake · blended ≥\$0.68/m ³ · brine permit

The water stack — line-by-line, with each credit's failure mode

The \$0.347/m³ audit: the base is measured anchors; each credit is a loop-internal yield with a named failure mode; the degraded cascade still clears the \$0.68 gate.

LINE	\$/M ³	PROVENANCE	WHAT MAKES IT FAIL
Base — solar energy 4 kWh/m ³ at the loop's own PPA + capex + O&M	\$0.567	MEASURED anchors (Rc13/Rw2/Rw3)	holds — if even the PPA lapses, grid power takes the base to \$0.88, still under SONEDE
DC reject heat → RO feed preheat — one seawater pipe cools the servers, then arrives warm at the membranes (SEC −2.5%/°C)	−\$0.007	DERIVED — Leg 03 + Leg 01, same pipework	fails only if no compute anchor tenant — and costs \$0.007
Kiln/exotherm preheat — the acid plant and board kilns warm the same feed	−\$0.000	DERIVED — Leg 04/05 waste heat	negligible by design — carried at zero margin
Brine valorization — salt/Mg evaporated on the loop's own free heat, sold at the park gate (of-take signed; nothing hauled)	−\$0.137	DERIVED — brine is the desal's own by-product	if the of-take is not signed by FID, the credit degrades to −\$0.03 (disposal avoided only) — the biggest single sensitivity
Shared civil works — one intake, one outfall, one grid connection serves RO + DC + metals in the same park	−\$0.032	DERIVED — 12% capex sharing	if the park phases one module at a time, halve it to −\$0.016
Dispatch value — the desal's own flexibility follows the loop's own solar curve	−\$0.015	DERIVED — 20 GWh shifted	fails to zero if the STEG contract does not reward flexibility — contractual, not hoped
ESG financing delta — the health leg's public KPIs reprice the loop's own debt (100–150bp)	−\$0.026	DERIVED — nations_v3	fails to zero if lenders do not accept the KPI track record — bankable-region fallback keeps the module alive
Brine pump-as-turbine — the desal's own residual pressure	−\$0.002	DERIVED — 2.5 bar × 75%	fails to zero — immaterial
Integrated	\$0.347/m³	28% of SONEDE's \$1.00–1.50 (Pg16)	the degraded cascade: if every soft credit fails, water still lands at \$0.521/m³ — under SONEDE's own \$1.00–1.50, and the \$0.68 gate still clears with margin. The credits are the margin, not the plan.

The people's energy is electrons, not hydrogen — and the H₂ lives inside the loop

The loop's H₂ (2.63 kt/yr) is an industrial molecule: per kWh of heat, green H₂ at \$0.21 is 7× the loop's \$0.031/kWh electron; blending caps at 5-20% by volume (Pg20/Pg21/Pg22); the whole H₂ volume equals 0.0% of grid-gas demand. The molecule goes where only a molecule works: ammonia for the GCT chain, inside the loop. the electrolyzer sits on the desal line, and everything it touches stays in the loop: the water it splits is the SWRO permeate (0.04% of the 150,000 m³/d output); its un-split reject stream is clean water that returns to the product line; its oxygen by-product aerates the spirulina ponds and greenhouses; its stack heat (15–20% of input) joins the kiln and DC reject heat on the desal preheat circuit; and the minerals concentrated in its flush stream join the brine valorization line. The electrolyzer is not a consumer of the loop — it is another recovery point inside it.

3. The capital stack and the ask

Phase 0: \$150-250k, 90 days, no construction — the price of replacing every projection with a measurement (itemized in the complete audit). **Then, module by module, on public gates:** \$58k health pilot · \$100M metals · \$550M solar · \$165M desal · \$5-30M truffles · \$2-5M food. Security per module: offtake assignment, STEG PPA, municipal contracts, MIGA/DFI channel — modeled, not committed.

What moves “commercially validated” from zero

GCT MOU + first assay batch · STEG PPA award · desal LOI ≥60% at ≥\$0.68/m³ · first truffle fruit + ≥€20/kg contract · 100-point water panel + 90-day health KPIs · 3,000-family cohort baseline. Each is a Phase-0/1 deliverable with a date.

What this document is: a concept-level engineering design whose numbers are computed in code from sourced primitives — recomputed, corrected, and applied into the design. Free and public.

What it is not: a commissioned state program, a bank-grade feasibility study, an investment offer, or a claim that any of this has been built. Nothing here is advice to any government or any investor. Nothing replaces geological, radiological, geotechnical or EPC-level surveys. Verify every figure and citation before relying on any of it.

Works cited

- [Rm8] 2024-2026 oxide prices: NdPr \$64-103/kg; Dy ~\$340/kg; Tb ~\$1,800-1,875/kg; La \$3.70/kg; Y <\$5-10/kg; Ce ~zero — <https://en.institut-seltene-erden.de/aktuelle-preise-von-seltenen-erden/...>
- [Rm10] Aclara: solvent-extraction separation plant Class-5 CAPEX \$244M, OPEX \$12/kg REO — [https://investingnews.com/aclara-announce-update-on-its-rare-earths-separation-project/...](https://investingnews.com/aclara-announce-update-on-its-rare-earths-separation-project/)
- [Rm15] Breakeven economics for non-Chinese REE projects (~\$77/kg NdPr-equivalent; PG-grade requires >\$150/kg) — <https://www.elibrary.imf.org/view/journals/001/2026/072/article-A001-en.xml...>
- [Rm16] Hydrochloric acid prices: EU \$124-150/t (Q4 2025); NA ~\$350/t; China ~\$15/t — <https://chemtradeasia.sg/en/market-insights/hydrochloric-acid-supply-report-2024-2025-buyer-lessons...>
- [Rw2] SWRO CAPEX \$800-1,200 per m³/day for mega-plants (>100k m³/d) — <https://supra-international.com/insights/seawater-reverse-osmosis-swro-desalination-technology-comprehensive-a...>
- [Rw9] 24/7 firming: 3.03 GW @20% CF = 0.606 GW avg (zero margin); 9.15 GWh/night; BESS round-trip 85% → 12-15 GWh; \$110-125/kWh → ~\$1.5B; solar+storage LCOE \$100-130/MWh — <https://www.energy-storage.news/battery-storage-system-prices-continue-to-fall-sharply-bnef-and-ember-reports-...>
- [Rw13] Grid constraints: 3 GW overwhelms local 400 kV; new lines \$1-2.5M/km; renewables <5% of generation; curtailment risk — <https://reglobal.org/tunisia-focuses-on-grid-expansion-for-integrating-renewable-energy/...>
- [Rc11] Tunisia sovereign ratings: Moody's Caa1 (Feb 2025 upgrade, stable); Fitch B- (Sep 2025, affirmed Jan 2026); reserves ~4.5 months import cover; WACC 10-15% — <https://www.fitchratings.com/research/sovereigns/fitch-upgrades-tunisia-to-b-outlook-stable-12-09-2025...>
- [Rc13] Empirical IPP record: 500 MW solar awarded 2024/25 (Scatec, Qair, Voltalia 25-yr PPAs at ~\$0.031/kWh); installed renewables ~400 MW; STEG debt >\$1.3B — <https://www.engineeringnews.co.za/article/scatec-awarded-ppas-for-wind-solar-projects-in-tunisia-2026-01-2>

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10. [Ru1] Uranium from wet-process phosphoric acid: Freeport (US) produced 14M lbs U₃O₈ 1978-99 at >95% DEHPA/TOPO recovery; ~20% of US uranium in the 1980s came from phosphates — <https://www.osti.gov/servlets/purl/1357599...>
11. [Ru2] URANEXT/OCP (Morocco) building the first modern yellowcake-from-phosphoric-acid plant, ~\$100M CAPEX, El Jadida, 2025 — <https://northafricapost.com/87963-uranext-to-build-moroccos-first-yellowcake-uranium-plant-in-el-jadida.html...>
12. [Ru3] WPA uranium SX cost \$59-83/lb U₃O₈ (IX \$46-76) vs \$80-95/lb 2024-26 spot — <https://repositories.lib.utexas.edu/bitstreams/89897275-85cf-4cd6-a0c5-1d8daa353799/download...>